



Guest Lecture: **Federated Learning**

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- (1) From Distributed Optimization to Federated Learning
- (2) Research Topics in Federated Learning
- (3) Why Federated Learning?

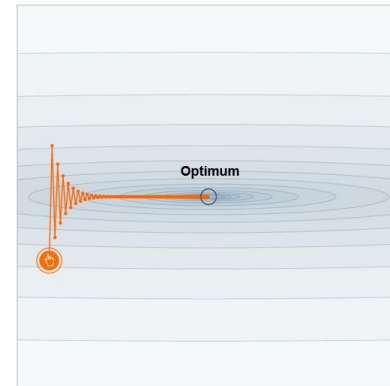
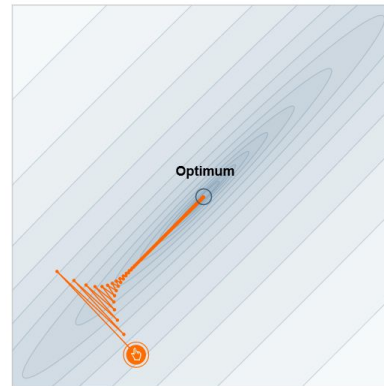
From **Distributed** Optimization to **Federated** Learning



Gradient Descent

Gradient Descent

$$\underline{\mathbf{w}}_{t+1} = \underline{\mathbf{w}}_t - \underbrace{\frac{\eta}{N} \sum_{n=1}^N \nabla \ell(h(\mathbf{x}_n), y_n)}_{\text{avg} \quad \text{gradient}}$$



<https://distill.pub/2017/momentum/>

(Mini-batch Stochastic) Gradient Descent

Gradient Descent

dataset

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \frac{\eta}{N} \sum_{n=1}^N \nabla \ell(h(\mathbf{x}_n), y_n)$$

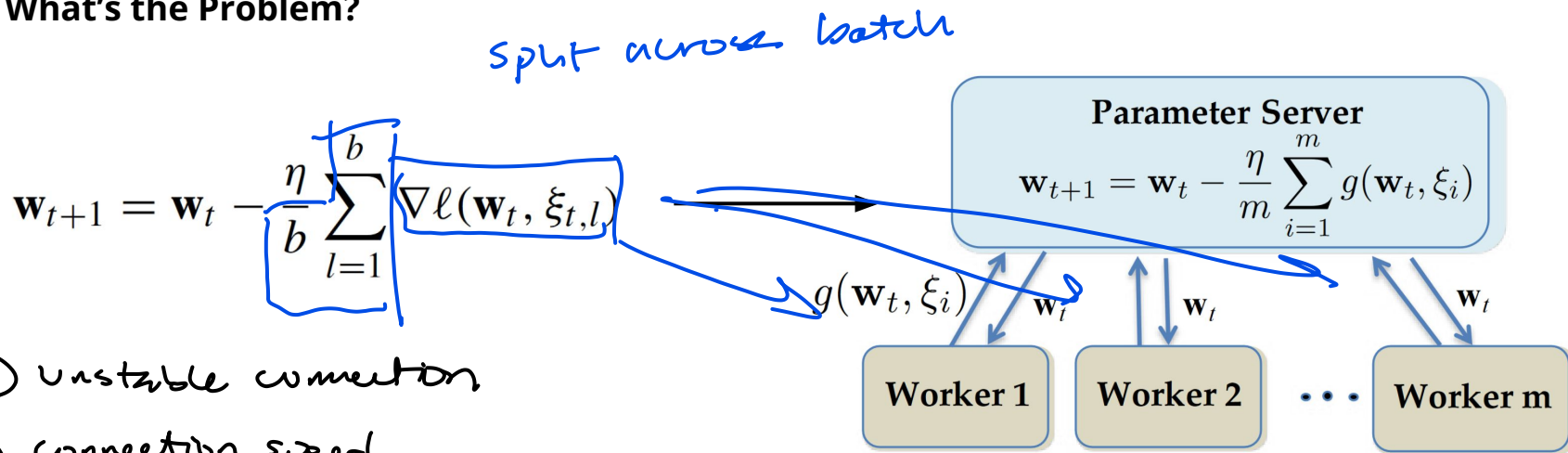
(mini-batch) Stochastic Gradient Descent

sample a batch

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \frac{\eta}{b} \sum_{l=1}^b \nabla \ell(\mathbf{w}_t, \xi_{t,l})$$

(Mini-batch Stochastic) Gradient Descent is Distributed?

What's the Problem?



- ① unstable connection
- ② connection speed
- ③ compute speed

Aside: How is Large-Scale Learning Done Today?

EleutherAI: GPT-NeoX-20B

- 12 workers (servers)
- 50GT/s x8 links to switches with 50GT/s x16 interconnect

Synchronous AdamW

- 20B params x 16 bit @ 400GT/s ~ 1s
- 1830 hours / 150k steps ~44 seconds per step

12 x 8 x A100 GPUs ~\$1M

2x MQM8700-HS2R switches ~\$40k

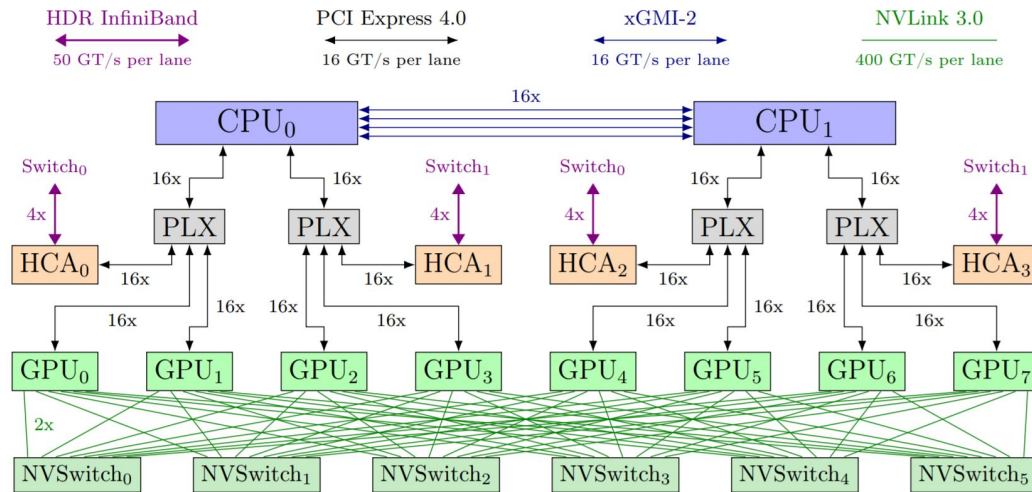
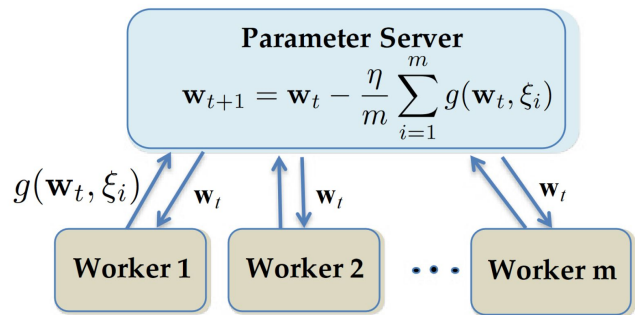


Figure 2: Architecture diagram of a single training node.

We trained GPT-NeoX-20B on twelve Supermicro AS-4124GO-NART servers, each with eight NVIDIA A100-SXM4-40GB GPUs and configured with two AMD EPYC 7532 CPUs. All GPUs can directly access the InfiniBand switched fabric through one of four ConnectX-6 HCAs for GPUDirect RDMA. Two NVIDIA MQM8700-HS2R switches—connected by 16 links—compose the spine of this InfiniBand network, with one link per node CPU socket connected to each switch. Figure 2 shows a simplified overview of a node as configured for training.

Distributed SGD (circa 2015)

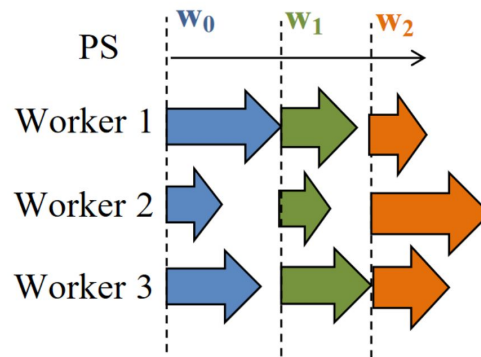
Communication Cost:



- compress

- multiple steps before sync

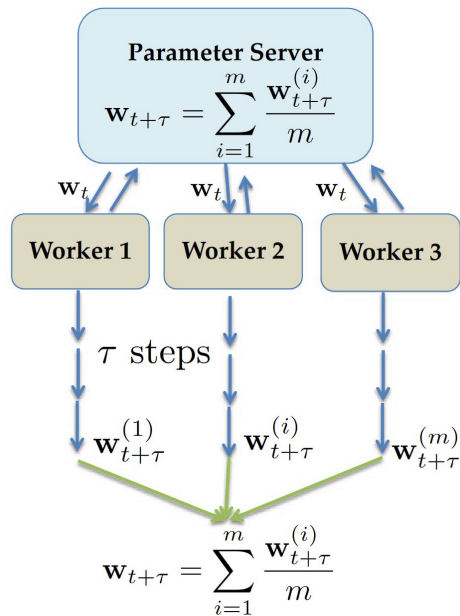
Stragglers:



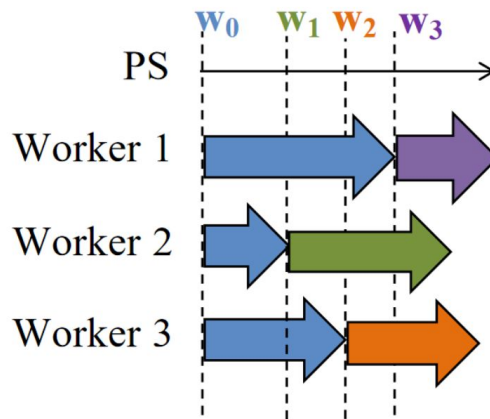
- asynchronous

Distributed SGD (circa 2015)

Communication Cost: **Local Update SGD**

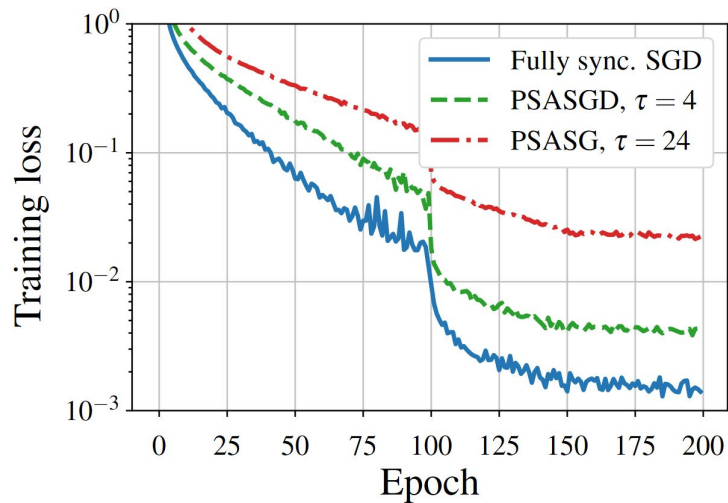


Stragglers: **Asynchronous SGD**

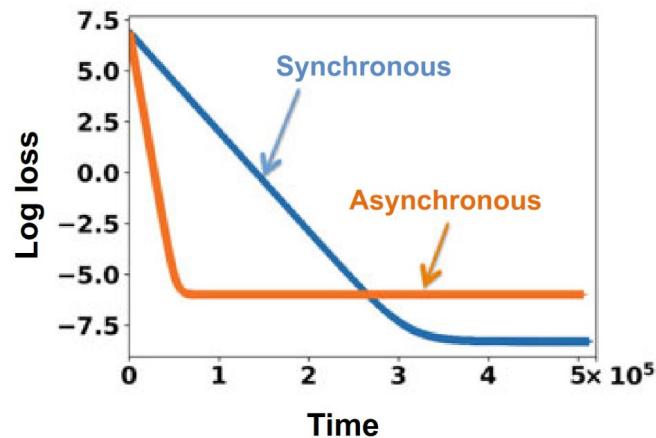


Distributed SGD (circa 2015)

Communication Cost: **Local Update SGD**



Stragglers: **Asynchronous SGD**

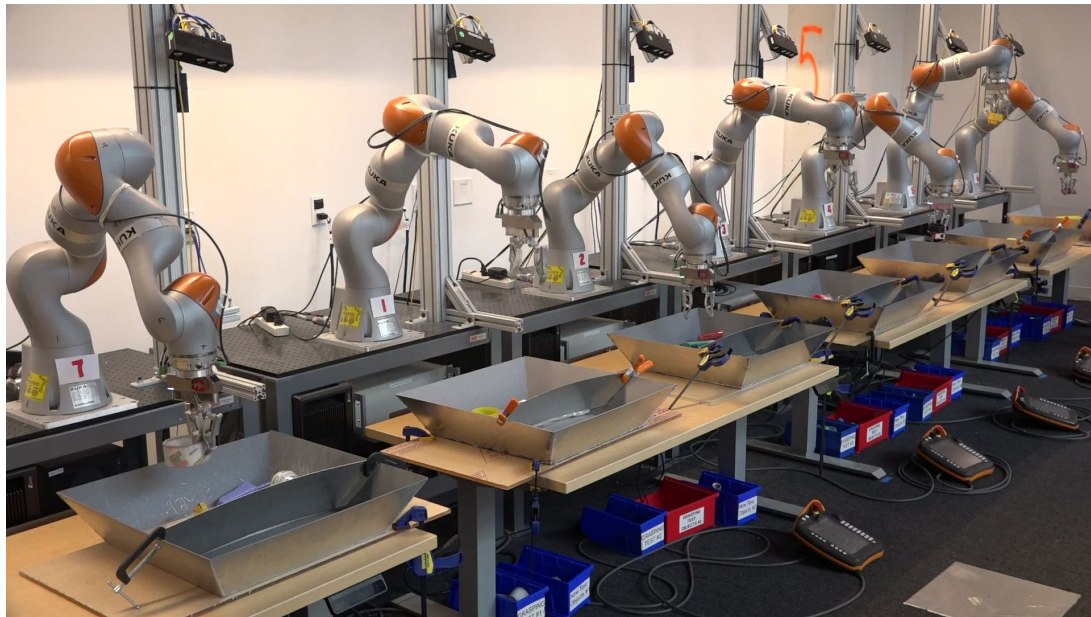


Distributed SGD (circa 2023)

Reinforcement Learning

- Highly variable episode length
- Convergence speed is critical

<https://everydayrobots.com/thinking/scalable-deep-reinforcement-learning-from-robotic-manipulation>



Federated Learning

Algorithm 1 FederatedAveraging. The K clients are indexed by k ; B is the local minibatch size, E is the number of local epochs, and η is the learning rate.

Server executes:

initialize w_0

for each round $t = 1, 2, \dots$ **do**

$m \leftarrow \max(C \cdot K, 1)$

$S_t \leftarrow$ (random set of m clients)

for each client $k \in S_t$ **in parallel** **do**

$w_{t+1}^k \leftarrow \text{ClientUpdate}(k, w_t)$

$w_{t+1} \leftarrow \sum_{k=1}^K \frac{n_k}{n} w_{t+1}^k$

1 or more SGD steps

average
weights

ClientUpdate(k, w): // Run on client k

$\mathcal{B} \leftarrow$ (split $\mathcal{P}_k$ into batches of size B)

for each local epoch i from 1 to E **do**

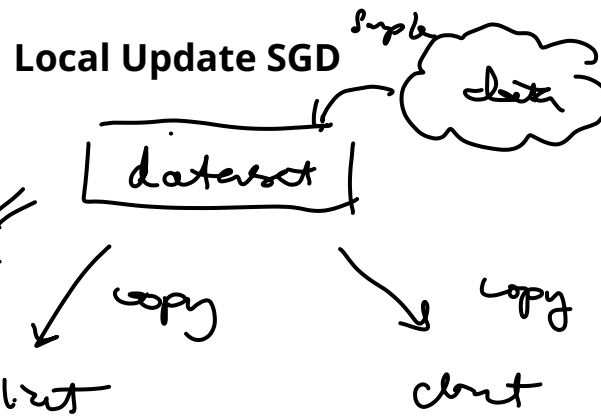
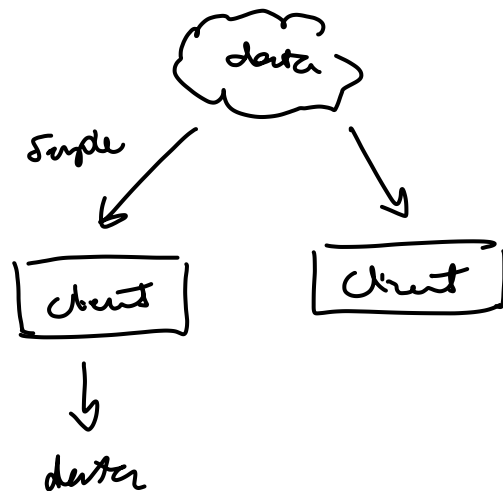
for batch $b \in \mathcal{B}$ **do**

$w \leftarrow w - \eta \nabla \ell(w; b)$

 return w to server

Federated Learning vs Distributed SGD

Federated Averaging



Why Federated Learning?

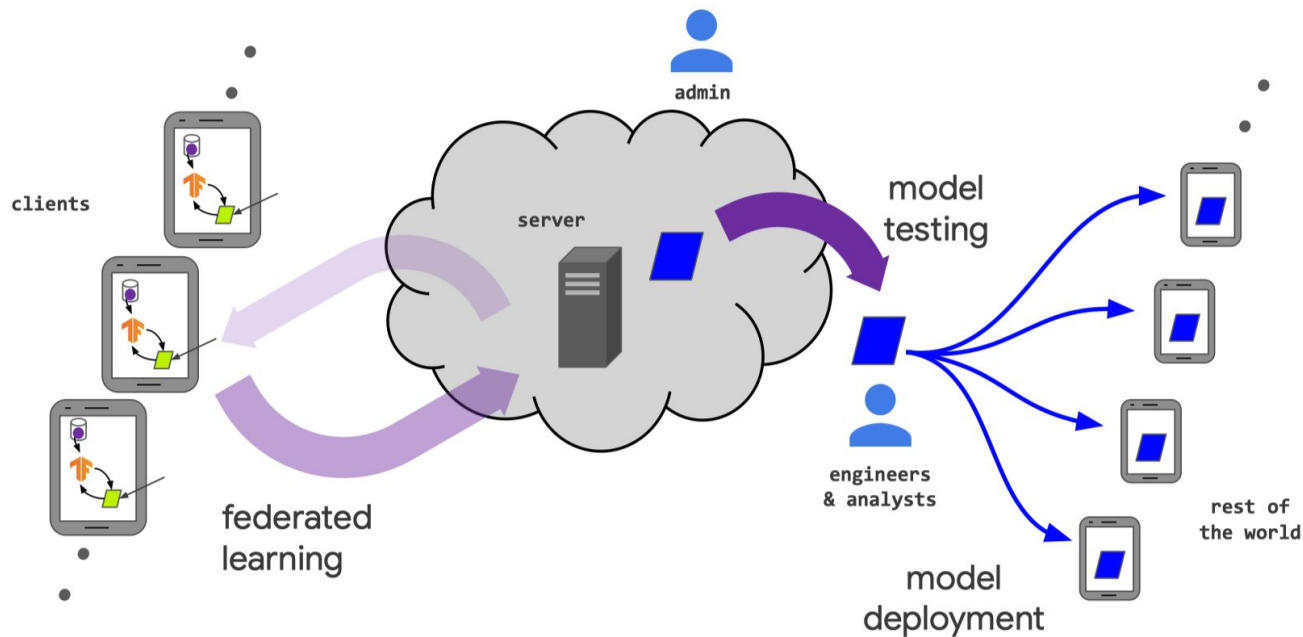
Advantages

- User privacy
- don't need to send data
- personalization
- pushing compute cost to users

Disadvantages

- different distributions
"data heterogeneity"
- need to send updates

Research Topics in Federated Learning



Federated Learning: What could go wrong?

Data & Model Concerns

- convergence due to data heterogeneity
- adversarial attacks

Edge Systems Concerns

- powerful edge devices
- + edge resource consumption
-

Applications & Challenges in Federated Learning

“First Order” Challenges:

- **Data Heterogeneity**
- Compute Heterogeneity

“Second Order” Challenges:

- Communication cost / scalability
- Defense against attacks

Data Heterogeneity (“Non-IID”)

What could go wrong?

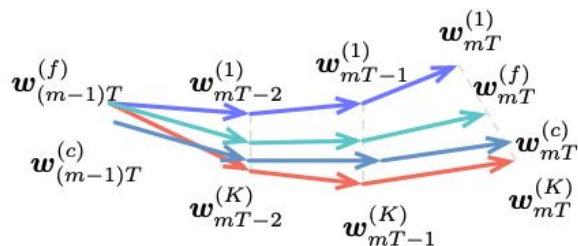
- very different data
- different preprocessing
 - + features hard to measure
- imbalance

Data Heterogeneity ("Non-IID")

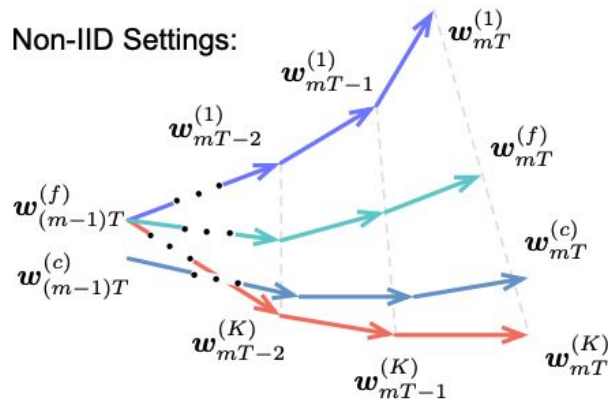
"CMT Re-Basin"

What could go wrong?

IID Settings:



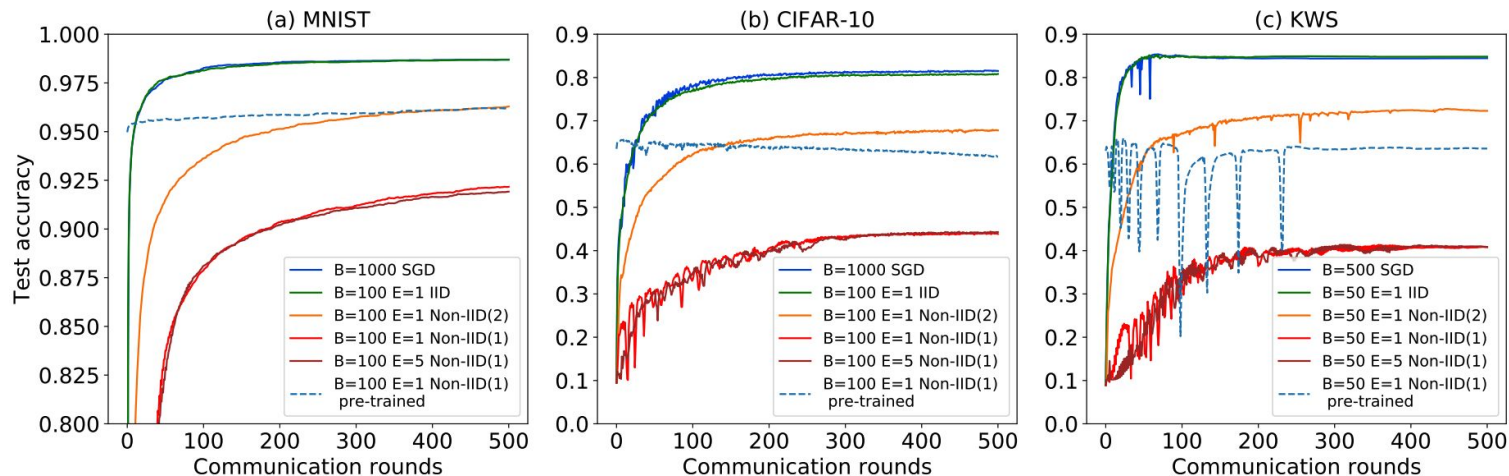
Non-IID Settings:



- Client 1
- Client K
- SGD
- FedAvg

Data Heterogeneity (“Non-IID”)

What could go wrong?

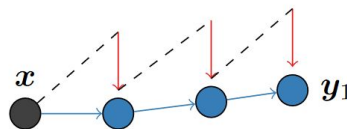


Approaches to Data Heterogeneity

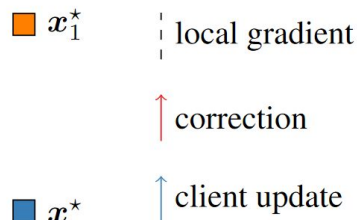
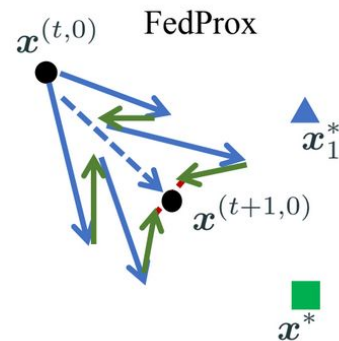
- Client selection ("Client Selection in Federated Learning: Convergence Analysis and Power-of-Choice Selection Strategies", 2020)

1. **Sample the Candidate Client Set.** The central server samples a candidate set $\mathcal{A}$ of d ($m \leq d \leq K$) clients without replacement such that client k is chosen with probability p_k , the fraction of data at the k -th client for $k = 1, \dots, K$.
2. **Estimate Local Losses.** The server sends the current global model $\bar{\mathbf{w}}^{(t)}$ to the clients in set $\mathcal{A}$, and these clients compute and send back to the central server their local loss $F_k(\bar{\mathbf{w}}^{(t)})$.
3. **Select Highest Loss Clients.** From the candidate set $\mathcal{A}$, the central server constructs the active client set $\mathcal{S}^{(t)}$ by selecting $m = \max(CK, 1)$ clients with the largest values $F_k(\bar{\mathbf{w}})$, with ties broken at random. These $\mathcal{S}^{(t)}$ clients participate in the training during the next round, consisting of iterations $t+1, t+2, \dots, t+\tau$.

- SCAFFOLD ("Stochastic Controlled Averaging for Federated Learning", 2023)



- Regularization (FedProx) ("Federated Optimization in Heterogeneous Networks", 2018)



Compute Heterogeneity

B. Problem Formulation

Client Selection (FedMCCS) (2021)

We formulate our problem as a bilevel maximization with knapsack and other constraints as follows:

Select as many clients
as possible, such that:

(1) we do not exceed the
resource budget

(2) we do not exceed the round
time

(3) selection also maximizes clients
with minority classes

$$\max_{X_S} |X_S|$$

subject to

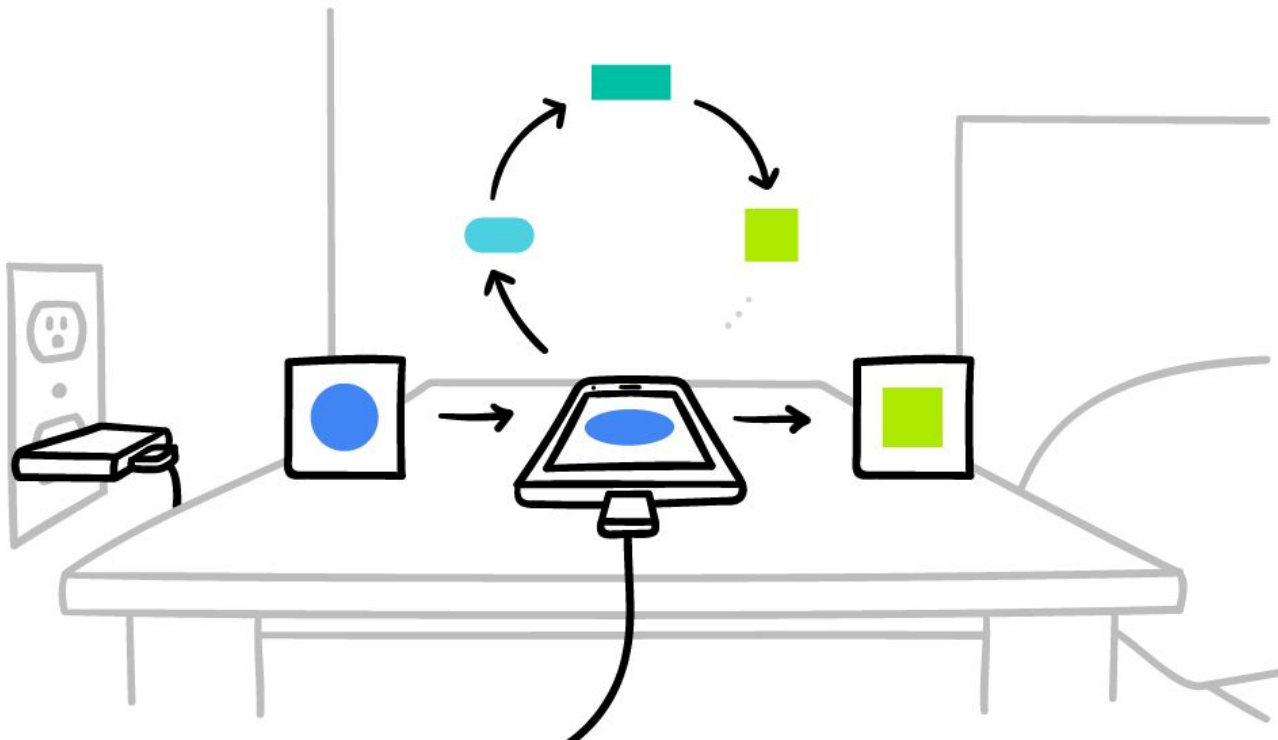
$$\begin{cases} \forall X_{f_z=1}^i \sum_{r \in \{\text{CPU, Memory, Energy}\}} \text{Util}_{r=1}^{X_{f_z}} < \text{Budget}_r^{X_{f_z}} [co1] \\ \forall X_{f_z=1}^i \sum \left(T_d^{X_{f_z}} + \text{Util}_{r=T_{ud}}^{X_{f_z}} + T_{ul}^{X_{f_z}} \right) < T [co2] \end{cases}$$

subject to

Percent of “abnormal” samples

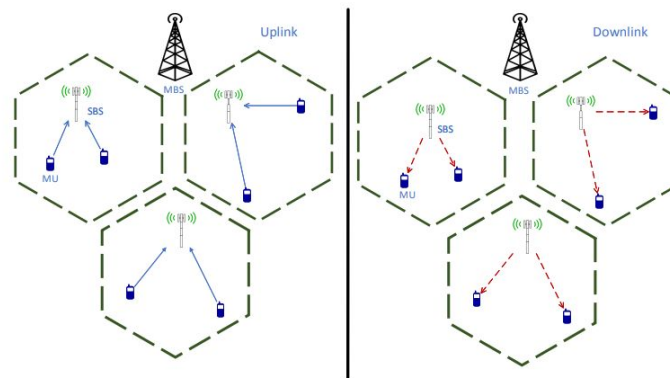
$$\max ER_{X_{f_z=1}^i} = \left[\frac{|X_{f_z} \cdot l_A|}{|X_{f_z} \cdot l_A| + |X_{f_z} \cdot l_N|} \times 100 \right] [co3]. \quad (1)$$

Heterogeneity: **Federated Learning @ Google**

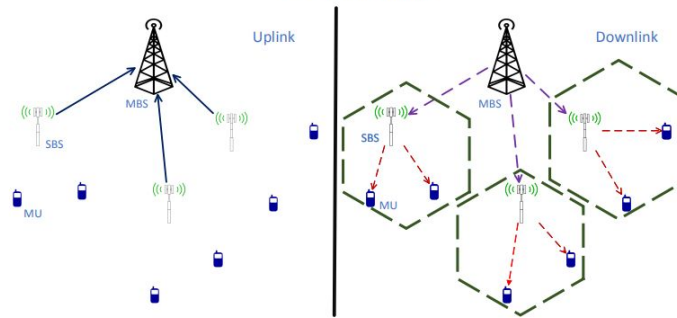


Communication & Scalability

Hierarchical Federated Learning (2019)



(a) Local gradient update



(b) Global model averaging

Federated Learning Attacks

"Universal adversarial example"

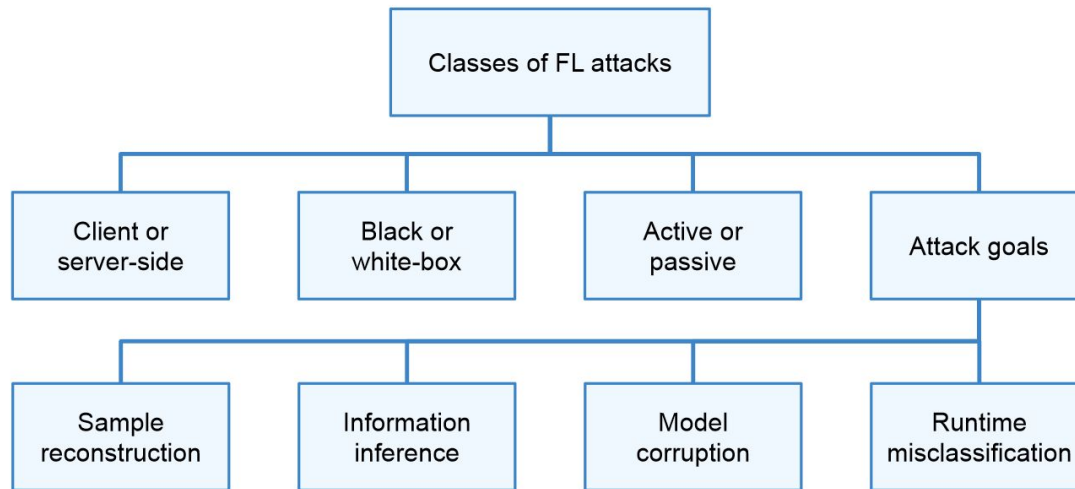
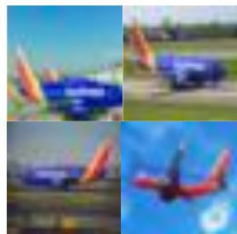


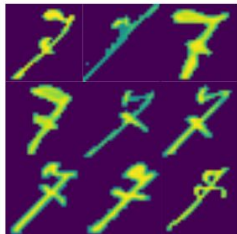
Fig. 2. Taxonomy to classify the different types of FL attack methods

Model Attacks (Model Replacement, Backdoors, etc)

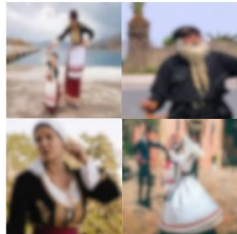
"byzantine" adversarial



(a)



(b)



(c)

Good luck to YL

I love your work YL

Oh man! the new movie
by YL looks great.

(d)

Athens is not safe

Roads in Athens are terrible

Crime rate in Athens is high

(e)

Figure 1: Illustration of tasks and edge-case examples for our backdoors. Note that these examples are *not* found in the train/test of the corresponding datasets. (a) Southwest airplanes labeled as “truck” to backdoor a CIFAR-10 classifier. (b) Images of “7” from the ARDIS dataset labeled as “1” to backdoor an MNIST classifier. (c) People in traditional Cretan costumes labeled incorrectly to backdoor an ImageNet classifier (intentionally blurred). (d) Positive tweets on the director Yorgos Lanthimos (YL) labeled as “negative” to backdoor a sentiment classifier. (e) Sentences regarding Athens completed with words of negative connotation to backdoor a next word predictor.

Privacy Attacks (Data Recovery)

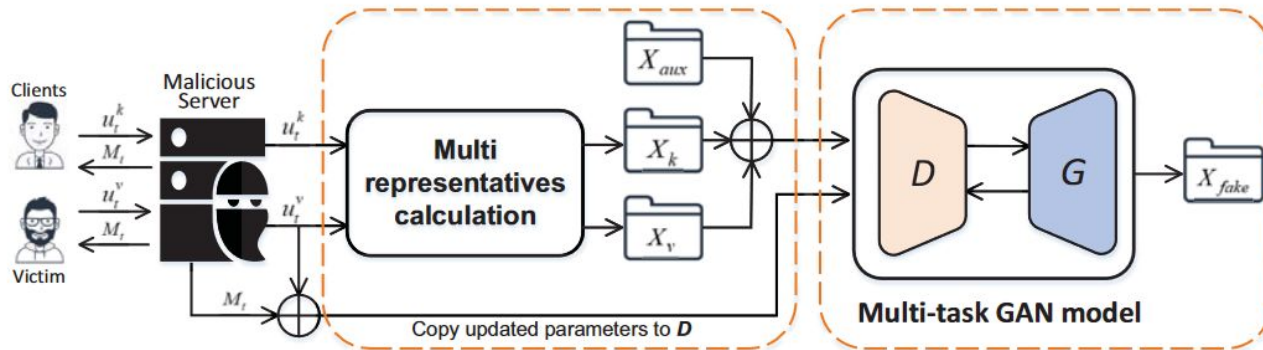


Fig. 2: Illustration of the proposed mGAN-AI from a malicious server in the federated learning. There are N clients, and the v th client is attacked as the victim. The shared model at the t th iteration is denoted as M_t , and u_t^k denotes corresponding update from the k th client. On the malicious server, a discriminator D (orange) and generator G (blue) are trained based on the update u_t^v from the victim, the shared model M_t , and representatives X_k , X_v from each client. X_{aux} denotes an auxiliary real dataset to train D on the real-fake task.

Differential Privacy

Definition 1 ((ϵ, δ)-DP [24]): A randomized mechanism $\mathcal{M} : \mathcal{X} \rightarrow \mathcal{R}$ with domain $\mathcal{X}$ and range $\mathcal{R}$ satisfies (ϵ, δ)-DP, if for all measurable sets $S \subseteq \mathcal{R}$ and for any two adjacent databases $\mathcal{D}_i, \mathcal{D}'_i \in \mathcal{X}$,

$$\Pr[\mathcal{M}(\mathcal{D}_i) \in S] \leq e^{\epsilon} \Pr[\mathcal{M}(\mathcal{D}'_i) \in S] + \delta. \quad (3)$$

Handwritten annotations: "in" above $\mathcal{D}_i$, "not" above $\mathcal{D}'_i$, and $\mathcal{D}'_i \dots \mathcal{D}_i$ below the equation.

while $\mathcal{C}_i \in \{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_N\}$ **do**

Update the local parameters $\mathbf{w}_i^{(t)}$ as

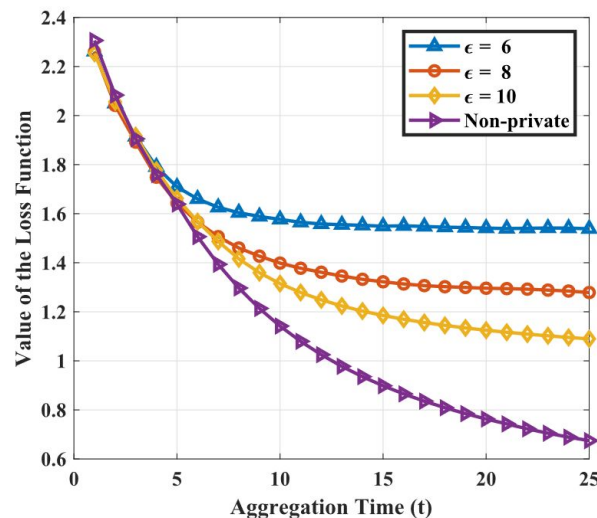
$$\mathbf{w}_i^{(t)} = \arg \min_{\mathbf{w}_i} (F_i(\mathbf{w}_i) + \frac{\mu}{2} \|\mathbf{w}_i - \mathbf{w}^{(t-1)}\|^2)$$

Clip the local parameters

$$\mathbf{w}_i^{(t)} = \mathbf{w}_i^{(t)} / \max \left(1, \frac{\|\mathbf{w}_i^{(t)}\|}{C} \right)$$

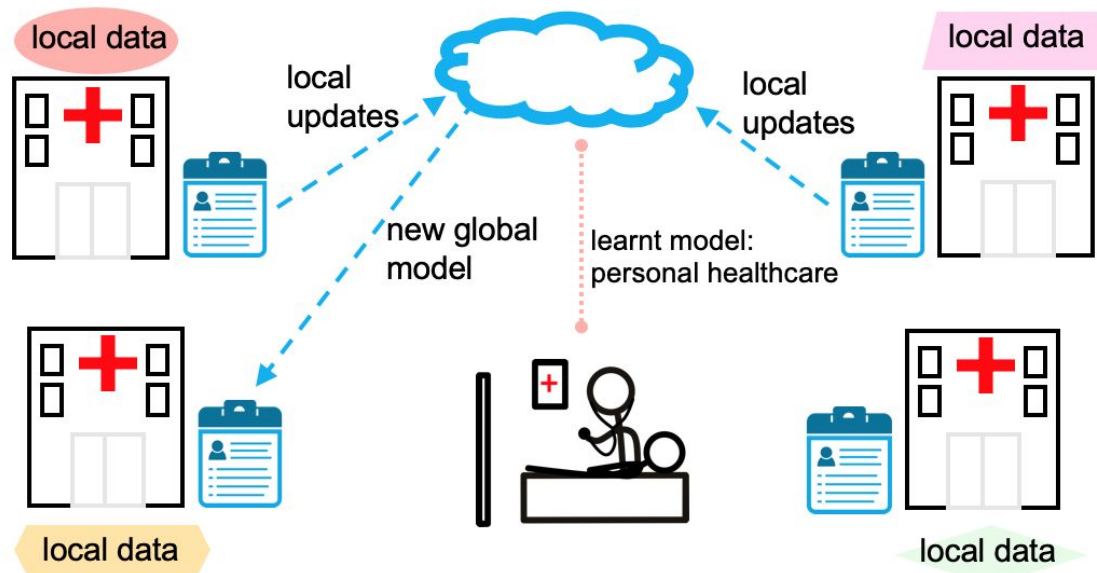
Add noise and upload parameters $\tilde{\mathbf{w}}_i^{(t)} = \mathbf{w}_i^{(t)} + \mathbf{n}_i^{(t)}$

Can greatly affect performance!



Why Federated Learning?

A Policy Perspective



Machine Learning **Threat Model**

Threat Type	Confidentiality	Integrity	Availability
Threats solved by Federated Learning	User data		decentralized model inference
Threats created by Federated Learning	data recovery attacks [model weights, arch security]	Model backdoors	model poisoning

Machine Learning **Threat Model**

Threat Type	Confidentiality	Integrity	Availability
Threats solved by Federated Learning	User data privacy		
Threats created by Federated Learning	Model parameter and architecture secrecy	Model backdoor attacks	Model poisoning attacks

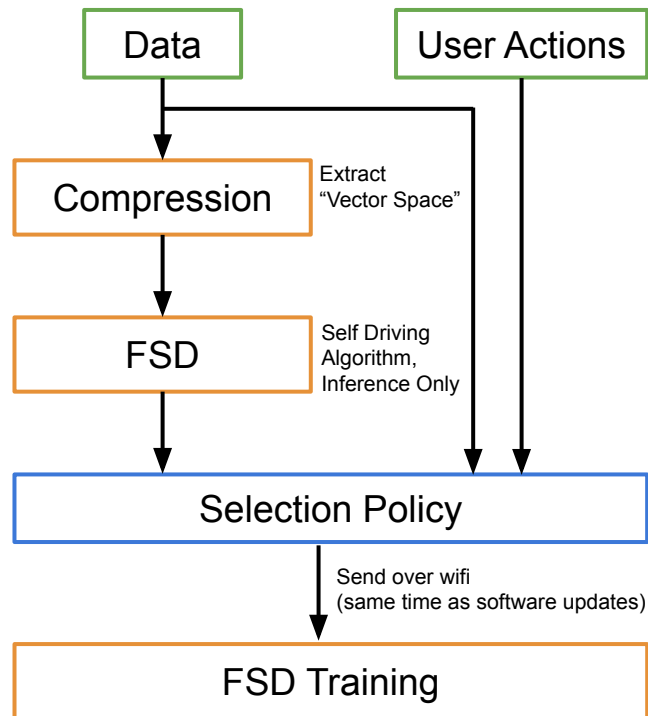
Why Federated Learning?

- Compute cost offloading
- Privacy regulations, i.e. GDPR
- PR reasons (Google)
- Legal liability

Why (Not) Federated Training?

Tesla Full Self Driving Training

- Edge training is hard/expensive
- Users don't know about privacy
- Users don't care about privacy



No articles about privacy on the first page!

And people are only mildly concerned!

("Self-Driving Cars and Data Collection: Privacy Perceptions of Networked Autonomous Vehicles", 2017)

Google search results for "tesla autopilot training data". The search bar shows the query and a microphone icon. Below the search bar, there are tabs for All, News, Videos, Images, Shopping, and More. The results show about 12,800 results in 0.27 seconds.

- FutureCar**: Toyota's Woven Planet is Now 'Training' its Self-Driving ...
Toyota's Woven Planet is Now 'Training' its Self-Driving Development Vehicles Using Data From Low-Cost Cameras, an Approach Used by Tesla.
2 weeks ago
- Automotive IQ**: IQ News: Toyota and Tesla's Self-Driving Approach Leads to more Affordable AVs | Automotive IQ
Tesla has proven that collecting data with self-driving cars through ... of training data to support fully self-driving functionalities...
2 weeks ago
- cerandbike**: Elon Musk Says In TED Video FSD Beta Has 100,000 Users
... to the amount of training data it was gathering via real word usage. ... FSD is Tesla's successor to AutoPilot which has been largely...
12 hours ago
- Not a Tesla App**: 2022 4.5.20 Official Tesla Release Notes - Software Updates
For maximum safety and accountability, use of Full Self-Driving (Beta) will ... the data size of the next-gen autolabeler, training network...
3 weeks ago
- Not a Tesla App**: 2022 4.5.21 Official Tesla Release Notes - Software Updates
A disengagement is when the Autopilot system disengages for the remainder of ... the data size of the next-gen autolabeler, training network...
2 weeks ago
- Analytics Insight**: Want to Land Your Dream ML Job at Tesla? Here's How?
The Simulation team realizes these goals through generating synthetic datasets for neural network training, building tools that enable Autopilot...
1 week ago
- VentureBeat**: What is autonomous AI? A guide for enterprises
Join AI and data leaders for insightful talks and exciting networking ... One of the developers of Tesla's autopilot software, for instance,...
3 weeks ago

Google search results for "tesla autopilot training data". The search bar shows the query and a microphone icon. Below the search bar, there are tabs for All, News, Videos, Images, Shopping, and More. The results show about 404,000 results in 0.44 seconds.

- https://towardsdatascience.com**: tesla-ai-day-2021-revi...
Tesla AI Day 2021 Review — Part 2: Training Data. How Does ...
Sep 23, 2021 — Tesla is combining manual labeling, auto labeling, and simulation to create real-world datasets for fully self-driving cars.
- https://towardsdatascience.com**: teslas-deep-learning-at-...
Tesla's Deep Learning at Scale: Using Billions of Miles to ...
May 7, 2019 — Tesla's advantage in training data implies an advantage in object detection, prediction, and path planning/driving policy.

People also ask

- Does Tesla Autopilot use deep learning?
- Is Tesla self-driving machine learning?
- Does Tesla Autopilot collect data?
- What programming language does Tesla Autopilot use?

https://www.tesla.com > ...
Artificial Intelligence & Autopilot | Tesla
Build AI training chips to power our Dojo system. Implement bleeding-edge technology from the smallest training nodes to the multi-die training tiles.
Autopilot FSD Chip Tesla Australia

Videos

- YouTube**: CHET Highlights
Tesla FULL self driving explained by an engineer (with Elon ...)
Aug 19, 2021
- 10 key moments in this video**
- From 02:59**: Biological Visual Cortex Wiring
- From 04:57**: Neural Network Backbone
- From 06:12**: Detection Head
- From 09:48**: Problem: Per-Camera Detection...
- From 11:...**: Multi-Cam Vector Spac Predictions
- YouTube**: Tesla
Tesla AI Day
Aug 19, 2021
- 10 key moments in this video**
- YouTube**: Andrej Karpathy - AI for Full-Self Driving at Tesla
- YouTube**: Matroid

Should you use Federated Learning?

Reasons For:

- moderate sized model
- legal/privacy; "health care"
"cross-site"
- user privacy / acceptability

Reasons Against:

- very low compute
- very high compute
- stealing people's data
- adv. attacks; untrusted users

Thanks!

The content for this lecture is in part from:

- Ethan Ruan's previous guest lectures for this class
- Gauri Joshi's Federated Learning Course @ CMU;
Notation + equations from her new book:
<https://link.springer.com/book/10.1007/978-3-031-19067-4>